

Mackenzie Hanh Cramer

Fun Fact: I am on in the top 1% of players on the 2048 global leaderboard and have crocheted 4 blankets
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EDUCATION

University of California, Berkeley

GPA 3.72

Berkeley, California

August 2021 - December 2024

- BA in Cognitive Science, Honors Student
- BA in Data Science, Domain Emphasis in Cognition
- Semester Abroad Spring 2024 at Yonsei University in Seoul, South Korea

WORK EXPERIENCE

Reader

Course: Natural Language Processing

Berkeley, California

January 2025 - Present

- Creating homework and coding assignments in Python
- Communicated during office hours for students and offered feedback towards questions concerning Natural Language Processing (NLP), assignments, and course information
- Formulated organizational system using Google Suite, specifically Google Sheets

Bancroft Library Student Worker

Under Iris Donovan

Berkeley, California

July 2022 - Present

- Worked with programs ALMA and AEON in tracking and identifying patron requests
- Leveraged Excel, Microsoft Excel spreadsheets and Microsoft Word, for data entry and reports
- Handling materials in UC Berkeley archives - paging, barcoding, retrieving, reshelving, and organizing various different items
- Trained 5+ new employees and was responsible for scheduling personal work shifts

Undergraduate Research Assistant

Under David Bamman, NLP Research

Berkeley, California

February 2023 - December 2024

- Applied NLP techniques in Python to analyze datasets, identify patterns and new insights within the fields of digital humanities and cultural analytics
- Annotated various datasets, scrutinized data guidelines to increase clarity and reproducibility
- Engineered visualizations and conducted analyses to monitor performance trends and identify patterns

UCEAP Intern

Under Hwanho Noh at Barun ICT

Seoul, Korea

March 2024 - June 2024

- Researched and pitched article ideas regarding data privacy and security
- 3 columns concerning subjects including AI chatbots modeled after deceased people, low-resource languages and lack of AI accessibility, and the privacy concerns of bioinformatics

Summer Data Analytics Intern

Under Stephen Zander at Evolent Health

Arlington, Virginia

June 2023 - August 2023

- Pioneered prediction model for time between biopsy and cancer treatment using Python
- Inspected insurance and membership claims using SQL, extracted Time Series measurements
- Collaborated with stakeholders such as health care experts, data scientists, and business professionals

Data Analytics Intern

Institute for Youth in Policy

United States

January 2023 - July 2023

- Produced data analytics for the data-education division, with focuses on identifying and measuring polarity of different political topics and potential improvements on high school curriculums
- Visualized data from workshops using Python and JASP for comprehensive statistical analysis.
- Gave feedback on others' visualizations and suggested improvements for clarity and minimized bias

Data Science Mentee*Undergraduate Laboratory of Data Science***Berkeley, California***September 2021 - June 2022*

- Aided in production of prediction model for number of COVID-19 cases in the US by sourcing data and clean data to extract trends
- Wrote, produced, and organized poster that we presented at poster convention at conclusion of the program and semester

Monthly Columnist*Under Melissa Dowling at The Kenwood Press***Kenwood, California***October 2020 - April 2021*

- Wrote 7 columns concerning education and the effect of political and social movements on it
- Responsible for outlining, writing, and editing articles for submission and feedback before print
- Surveyed and gathered opinions from community members on articles' topics

PROJECTS**Honors Thesis: Analyzing Descriptions in Novels***Mackenzie Cramer***Berkeley, California***August 2024 - December 2024*

- Extracted characters information from 5,360 novels with Named Entity Recognition (NER) and NLP
- Mapped characters' descriptions onto their roles to identify common descriptions in varying genres
- Produced 20+ page report alongside visualizations and trend explanations

Youtube Apology Analysis*Mackenzie Cramer, Sabrina Baur***Berkeley, California***August 2023 - December 2023*

- Employed NLP techniques in order to analyze patterns that arise among Youtube Apology videos
- Used Python while employed NLTK, sequence alignment, coreference chains, and other techniques
- Sourced, collected, and cleaned dataset and produced visualizations and analyses of trends

Maria Carrillo Diversity Improvement Proposal*Mackenzie Cramer, Zoya Ahmed***Santa Rosa, California***July 2020 - December 2020*

- Worked with district officials, school staff, and students within the Santa Rosa City School District
- Developed surveys, 300+ responses, and researched history of policies in order to suggest new and effective ways to connect and help minority and underserved students
- Presented 90+ page proposal findings to teachers, administrators, and interested district officials

PUBLICATIONS**Speak, Memory: An Archeology of Books Known to Chat-GPT/GPT-4***Kent K. Chang, Mackenzie Cramer, Sandeep Soni, David Bamman**October 2023*<https://aclanthology.org/2023.emnlp-main.453/>

Abstract: In this work, we carry out a data archaeology to infer books that are known to ChatGPT and GPT-4 using a name cloze membership inference query. We find that OpenAI models have memorized a wide collection of copyrighted materials, and that the degree of memorization is tied to the frequency with which passages of those books appear on the web. The ability of these models to memorize an unknown set of books complicates assessments of measurement validity for cultural analytics by contaminating test data; we show that models perform much better on memorized books than on non-memorized books for downstream tasks. We argue that this supports a case for open models whose training data is known.

VOLUNTEERING**Ethnic Studies Curriculum Advisor***With Ethnic Studies Course Curriculum***Santa Rosa, California***January 2021- March 2021*

- Worked on committee focused on High School English Literature Ethnic Studies course, to be deployed by Santa Rosa City School District

- Pitched topics to teach, promoting texts and books to be taught to students
- Worked with teachers, other students, and aided in course development

SCHOLARSHIPS

Redwood Empire Chinese Association

Sonoma, California

March 2021

Awarded to graduating high school seniors of Asian descent for excellent academic performance and community service, with plans to pursue post-secondary education.

ADDITIONAL WORK

SMASH Panel

Berkeley, California

Hosted by Anoop Kaur and Maggie Melone-Echiburú

July 2024

Panel Member for SMASH (Summer Math And Science Honors) Academy at UC Berkeley Campus and discussed admissions, taking Computer Science courses, and advice for applying and networking

One Nation Together

Sonoma, California

Ep. 5: Zoya Ahmed and Mackenzie Cramer

November 2020

<https://open.spotify.com/episode/6AxvDxZB4j8huDkLw4KBCK>

Guest starred on One Nation Together Podcast hosted by Aarav, Idan, and Reynalda to discuss Universal Community for Change (UCC) and efforts to make our high school campus a safe space for individuals of all racial and sexual identities.